STRUCTURA: A Bayesian Network Framework for Information-Theoretic Scoring and Behavioral Anomaly Detection

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1 · INTRODUCTION

Under network psychometrics, constructs are modeled as interacting nodes linked by explicit dependencies. Bayesian Networks (BNs) enable exact person-level inference,[1,2,3] but their use in cognitive assessment remains limited.**No validated framework currently turns a single response pattern into a reliable, interpretable, and clinically meaningful score.**[4] STRUCTURA is a novel framework, based on BNs topology, specifically developed to address this gap.

2 · METHODS

1 **Reference sample & instrument.** Reference data: 66 kindergarten children (*M* age = 4.72, *SD* = 0.30; 46.9% male) from Orsoni and colleagues.[5] Raven's CPM, a non-verbal measure of fluid intelligence and analogical reasoning, includes 36 dichotomous items across subscales A, Ab, and B; after excluding zero-variance items, 26 items were analysed.

Compilation & local evidence. The BN was compiled into a junction tree for exact inference, with Laplace-smoothed CPTs (α = 0.5). For each item, local evidence was defined by its Markov Blanket: $E_i = \mathrm{MB}(X_i)$. Only structurally relevant neighbours informed person-level inference.

2 **Metrics.** Three complementary indices are derived:

TESS. $\sum_i P(X_i = 1 \mid E_i)$, normalised to 0–100. Equal raw accuracy can yield different TESS values.

Global SI (GSI). $\sum_i -\log_2 P(X_i = x_i \mid E_i)$, mapped onto an empirical 0–100 normative centile scale.

Local SI (LSI). $\mathrm{SI}_{\pm,\,i} = (x_i - E[X_i \mid E_i]) \cdot I(x_i)$: signed item-level anomaly; flagged at the $5^{th}/95^{th}$ normative percentiles.

Uncertainty. Parametric bootstrap (95% CI): each CPT perturbed via Dirichlet sampling (ISS as hyperparameter).

3A · RESULTS — TWO RESPONSE PROFILES

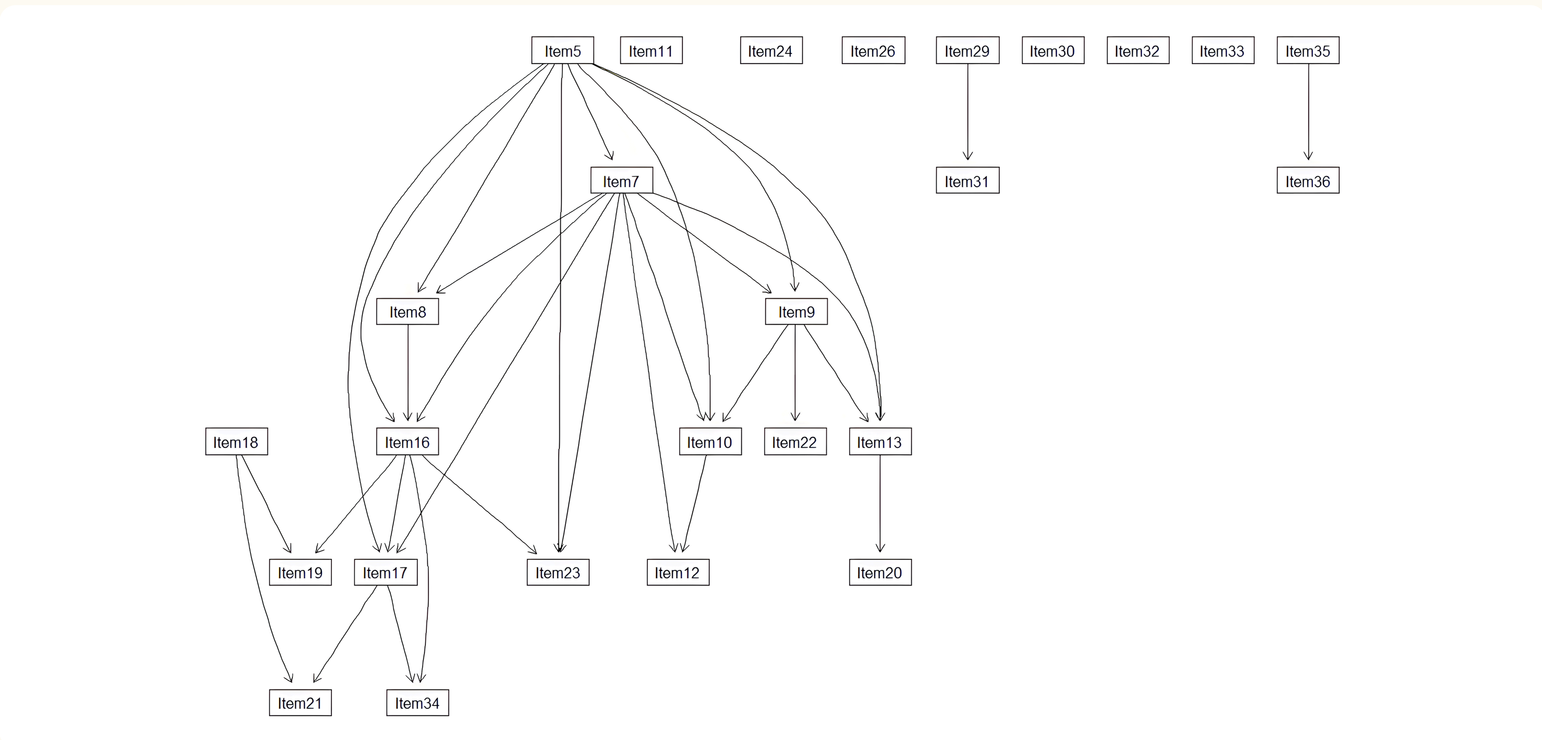

Fig. 1 — Learned normative BN of Raven's CPM, adapted from Orsoni and colleagues.[5]

CASE 1 — GUTTMAN PARADOX [6]

15/26 items correct: failing easy items while solving difficult ones.

57.7% - <5th RAW ACCURACY & NORMATIVE CENTILE	52.9% - <1st TESS CI 50–57 & NORMATIVE CENTILE	98.5th GSI - CI 97–100	3 + 7 CARELESS + LUCKY FLAGS

CASE 2 — NORMOTYPICAL PROFILE

22/26 items correct: empirical profile from reference sample.

84.6% - ~52th RAW ACCURACY & NORMATIVE CENTILE	73.0% - ~88th TESS CI 71–75 & NORMATIVE CENTILE	18.2th GSI - CI 0–55	0 + 1 CARELESS + LUCKY FLAGS

4 · DISCUSSION & CONCLUSIONS

These three metrics capture complementary aspects of anomaly:**TESS** adjusts the ability estimate for structural inconsistency,**GSI** quantifies how improbable the overall response pattern is, and **LSI** identifies the specific items driving the anomaly.

STRUCTURA translates BN topology into individual-level scoring.

Uncertainty is rigorously quantified via Dirichlet bootstrap CIs.

Generalizable to any assessment where a DAG is theoretically or empirically justified.

Next steps: Formally validate TESS, GSI, and LSI; integrating contextual nodes and evaluating clinical feasibility in psychiatric populations.

3B · RESULTS — LOCAL SI PROFILES

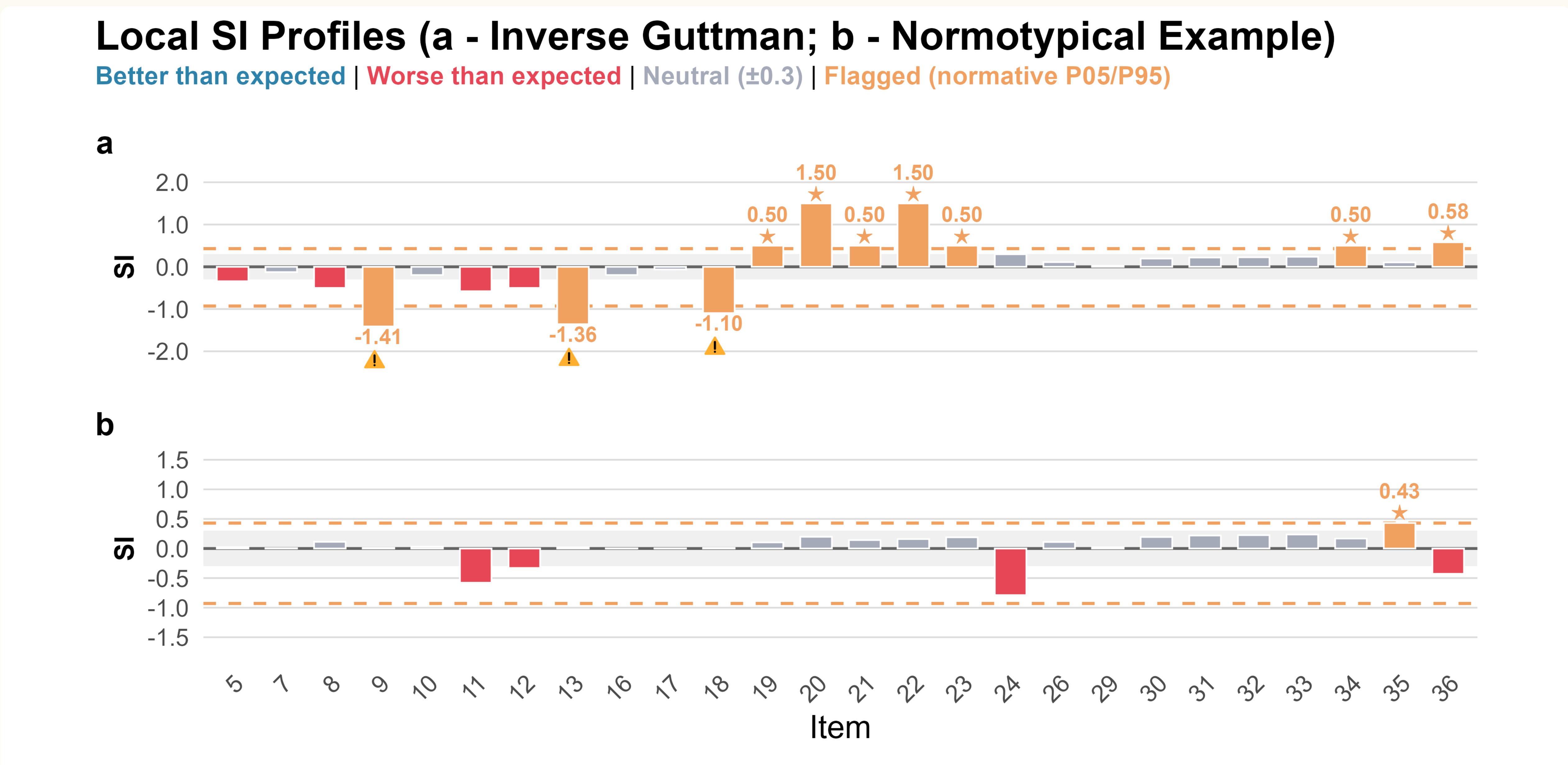

Fig. 2 — Signed SI per item. P05 = −0.929 and P95 = 0.430. a - Guttman profile; 3 careless and 7 lucky flags. b - Control profile; 0 careless and 1 lucky flag.

QR CODE

Scan for full references, and poster PDF.

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